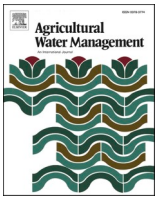
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Micro-irrigation adoption and the Jevons' Paradox: A study from four states of India

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ABSTRACT

Micro-irrigation, known for enhancing water use efficiency and crop yields, is subject to the phenomenon of the Jevons paradox or rebound effect. Farmers adopting the technology also tend to adopt economic mechanisms – extensive (increasing operated area) and intensive (changing crop patterns). We studied the question of the rebound effect using primary data from four states of India – Maharashtra, Telangana, Uttar Pradesh, and Madhya Pradesh. Recall data from 404 sample farmers were collected on crop area, land use, and water use in a pre- and post-micro-irrigation adoption scenario. The data analysis suggests that most farmers agreed they increased their cropped area after micro-irrigation. The recall data of area change under various crops showed that more than 85 % of farmers increased their area post the adoption of micro-irrigation. Along with the increase in total operated area, farmers brought an additional 267 ha (17 %) under irrigated agriculture, out of which about 87 ha were converted from rainfed to irrigated areas under various crops. Farmers converted a substantial area from cereals to high-value horticulture crops. Micro-irrigation was found to reduce pumping hours for irrigation (a proxy for volumetric water use); however, certain crops exhibit an absolute increase in water consumption. The technology, therefore, fulfils two key mandates of the policy *Pradhan Mantri Krishi Sinchayee Yojana* (Prime Minister's Agricultural Irrigation Scheme) - "per drop more crop" and "*Har Khet Ko Pani*" (water for every farmland), but at a cost of increased risk to future consumptive uses. The research advocates a more comprehensive basin-scale investigation of farmers' behavioural responses to the adoption of micro-irrigation and the subsequent changes in volumetric consumptive water use.

1. Introduction

Globally, water consumption has been steadily increasing, with an average annual increase of one percent over the past four decades (UN, 2020). As much as 80 percent of the population living under water stress resides in Asia, particularly in northeast China, India, and Pakistan. Projections indicate a steep increase in water scarcity among global urban populations by 2050, especially in India (He et al., 2021). The imminent water crisis poses a severe threat to food security and is driven by global megatrends such as urbanisation, climate change, rising middle-class populations, and inequality (He et al., 2021; King et al., 2017). Projections also suggest a staggering 40 percent gap between water demand and sustainable supply by 2030 if current trends persist, leading to economic strains similar to the COVID-19 slowdown (UN, 2020). In achieving Sustainable Development Goal (SDG 6) — "ensure availability and sustainable management of water and sanitation for all" —managing irrigation water demand, which accounts for about 70 percent of global water extractions, is critical (Grafton et al., 2018).

Many nations, including India, have responded to increasing water scarcity by implementing policies to reduce water demand in agriculture. These efforts have endorsed technologies like micro-irrigation, and infrastructure development that enhance irrigation efficiency (Wheeler et al., 2020). The Government of India has been promoting micro-irrigation with subsidies, under a policy called the *Pradhan Mantri Krishi Sinchayee Yojana* (Prime Minister's Agricultural Irrigation Scheme), renamed after merging several previously run schemes (Gandhi et al., 2021; GoI, 2017). Two major aims of the policy are "Per Drop More Crop" and "*Har Khet Ko Pani*" (water for every farmland).

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Micro-irrigation technology has shown to help farmers produce more from less inputs - water, fertilisers, labour, and energy, and thus enhance their incomes (Gandhi et al., 2021; Mattoussi et al., 2023; Narayanamoorthy, 2005; Scheierling et al., 2006a; Viswanathan et al., 2016). However, some argue that the technology, while enhancing private benefits, might not conserve water and could even expedite the depletion of water resources through the phenomenon of Jevons paradox, often called the rebound effect (Gómez and Pérez-Blanco, 2014; Pérez-Blanco et al., 2021; Richter et al., 2017).

The paradox may result in outcomes that are contrary to the intended goals of a policy (Alcott, 2005) designed to reduce water resource consumption. Some research even indicates an increase in groundwater demand following the adoption of micro-irrigation (Bahinipati and Viswanathan, 2016; Caswell et al., 1984; Fishman et al., 2023; Khanna and Zilberman, 2017; Narayanamoorthy, 2009; Pérez-Blanco et al., 2021). The conservation outcome depends on the elasticity of water use efficiency demand — that is, how much additional demand arises with a unit increase in water use efficiency. If demand is elastic to water use efficiency, then water consumption will increase (Caswell et al., 1984). The increase is due to farmers' behavioural responses to water savings and higher incomes from improved water efficiency. Various other studies also highlight this paradoxical situation and emphasise on the need to consider farmers' behaviour within different hydrological and regulatory environments (Dagnino and Ward, 2012; Lecina et al., 2010; Lopez-Gunn et al., 2012; Perry and Karajeh, 2017, 2017; Scheierling et al., 2006b; Ward and Pulido-Velazquez, 2008).

The literature and practice have widely demonstrated the benefits of micro-irrigation (MI) in Indian agriculture, but not much research is available examining the possible Jevons paradox phenomenon (Jevons, 1866) that micro-irrigation can have. Limited research is available examining this in-depth in a developing country context such as India. Given this background, the research has studied the impact of micro-irrigation in agriculture, and particularly the incidence of the Jevons paradox or rebound effect, based on primary surveys in four major states of India which have implemented the technology.

The main objective of this study is to comprehensively examine the behaviour around the paradox through a broader sample. This is achieved by covering four major states of India — Uttar Pradesh, Madhya Pradesh, Telangana, and Maharashtra — which have implemented MI. The study uses comprehensive primary survey recall data on crop cultivation pre and post adoption of the technology, examining related changes such as shifts to high water-consuming crops, increases in crop density, expansion of cultivated areas, and shifts to more water-intensive varieties. The study provides deeper insights into the economic mechanisms (Song et al., 2018) of the paradox.

The paper is structured as follows: In Section 2, we review the relevant literature on Jevon's paradox and its two mechanisms, economic and hydrological. In section three we outline the study's methodology and data collection strategy. In section four we present the analysis, including a description of the data, details on pre- and post-MI adoption changes in crop area and patterns, and farmers' perceptions regarding area increases. We use irrigation hours as a proxy to calculate differences in water use per unit area of crop and compute total irrigation hours for the farm. Regression analysis was employed to assess changes pre and post adoption of MI and their statistical significance. In section five we provide a detailed discussion of the key findings, emphasising their policy relevance and limitations. In the final section we have summarised the key conclusions.

2. Literature review

Water use efficiency can be enhanced through technologies such as micro-irrigation, mulching, cover crop. These efficiencies can be categorised as - technical, cost, and productivity (Gómez and Pérez-Blanco, 2014). Technical efficiency measures the effect of input use on output; cost efficiency assesses how a technology can save costs for a given output level; and productivity measures output obtained per unit of water. Although all are desirable, enhanced water productivity can lead to increased water consumption as farmers aim to boost profits through expanded cropping areas and altered patterns (Gómez and Pérez-Blanco, 2014; Fishman et al., 2023; Wheeler et al., 2020). The phenomenon - the rebound effect or Jevons paradox, is more pronounced among farmers with larger water rights, as they exercise their rights to over extract groundwater, leading to spatial externalities and water inequity that affect other farmers in the same basin (H. Li and Zhao, 2018). The paradox manifests through expanded cultivation areas, shifts to water-intensive crops, and increased water usage behaviours among farmers after the adoption of micro irrigation. A comprehensive understanding of these mechanisms is essential for effective policy formulation and mitigation strategies. Song et al. (2018) identified the paradox taking place through two mechanisms — hydrological and economic.

The hydrological mechanism encompasses changes in crop water consumption due to increased plant canopy, variations in evapotranspiration, and modifications in surface runoff influenced by land slope (Dagnino and Ward, 2012; Lecina et al., 2010; Song et al., 2018). Studies in regions such as New Mexico's Rio Grande Project (Brinegar and Ward, 2009) and the Upper Rio Grande basin (Dagnino and Ward, 2012) indicate that while subsidies for water-saving technologies enhance water use efficiency, they also increase crop water consumption through evapotranspiration. Higher crop growth and yields exacerbate water losses due to increased biological water demand, as evidenced in California's Westlands water district (Burt et al., 2001). Lopez-Gunn et al. (2012) observed similar trends in Spain, where efficiency gains from MI did not translate into actual water conservation due to increased evapotranspiration and overall water consumption in cropping systems. In another study in the Upper Rio Grande Basin, water efficiency improvements correlated positively with crop yield and subsequent additional evapotranspiration due to MI (Ward and Pulido-Velazquez, 2008).

The adoption of new agricultural technology by farmers is generally driven by economic incentives rather than solely by water conservation (Brinegar and Ward, 2009). Economic gains, such as reduced operational costs in irrigation and increased yields, steer farmers toward profit-maximising behaviours upon implementing micro-irrigation (MI) technology (Song et al., 2018). The behavioural shift includes altering cropping patterns, changing crop densities, expanding cultivation areas, and switching to more water-intensive crop varieties, reflecting farmers' prioritisation of economic outcomes over water conservation, termed the economic mechanism (Wheeler et al., 2020; Fishman et al., 2023; Song et al., 2018). The economic mechanism can be further divided into extensive and intensive economic mechanisms. The extensive mechanism involves expanding cultivated areas to increase output for specific crops, while intensive mechanism aims to achieve higher production from the same resources - land and water, through the adoption of hybrids or increased plant density. Studies reveal a pattern across various countries, including India, where increased water use efficiency leads farmers to expand cultivation areas — sometimes into low-quality fallow land, potentially offsetting water-saving benefits for economic gains (Bhamoriya and Mathew, 2014; Birkenholtz, 2017; Pfeiffer and Lin, 2014). Additionally, increased water use efficiency triggers crop intensification, as farmers shift from dryland to irrigated crops or change cropping patterns towards horticultural or water-intensive crops (Fishman et al., 2023). Studies corroborate this intensification phenomenon, showing an increase in water consumption ranging from 4 to 45 percent due to shifts towards water-intensive crops after MI technology adoption (González-Cebollada, 2015).

In Indian context, Bahinipati and Viswanathan (2019) conducted a study across agro-ecological areas in Gujarat state, analysing the impact of micro-irrigation on water conservation. Their findings suggested that micro-irrigation alone was insufficient for water conservation and proposed combining MI with metered electricity supply. Birkenholtz (2017) explored the effect of micro-irrigation through qualitative interviews

with 12 farmers and reported that farmers extracted equal (and sometimes more) groundwater after adopting MI as they increased the area under cultivation. The study by Patil et al. (2015) found that the adoption of micro-irrigation resulted in an increase in both gross-irrigated area and farmer income without a significant surge in groundwater use. Fishman et al. (2023) examined the impact of drip irrigation subsidies in Andhra Pradesh through a randomized control trial. Over three years, it found that farmers switched to more profitable horticulture crops, leading to increased profitability. The farmers sold the saved water to neighbouring farmers, resulting in no significant decline in groundwater extraction. However, these previous studies were limited in their geographical scope and context, and they did not offer comprehensive understanding of the paradox's dynamics. This limitation presented for us an opportunity to further explore the paradox and deepen our understanding. The economic and hydrological mechanisms post MI technology adoption can have unintended implications for government water conservation efforts. We primarily focused on analysing the economic mechanisms of the rebound effect in our study. Table 1 below summarises the research and findings on the paradox discussed above.

Table 1
Literature summary on rebound effect.

Key Highlights	Authors
The Paradox	
When technology efficiency gains, increases the consumption of the resource rebound effect occurs	(Jevons, 1866; Sorrell and Dimitropoulos, 2008)
Consumption of water will depend on water use efficiency elasticity of demand, if demand is elastic than water consumption will go up	(Caswell et al., 1984)
The withdrawal of water is not just dependent on the method or the usage of the technology but how a farmer, reacts to the improvement of water use and the additional quantity of water available for use consumption	(Gutierrez-Martin and Gomez, 2011)
Rebound effect occurs due to factors such as post adoption behaviour like cropping area expansion, open water rights, and absence of environmental goals for the farmers.	(Song et al., 2018)
With improvement of water productivity, i.e., output obtained per volume of water use, there exist a natural tendency of farmer to increase her profit by using saved water.	(Thiam et al., 2001)
The effect of the paradox is higher in case of the farmers those who have larger water rights, as the farmer tend to grow more water intensive crop, increase crop area, and intensify crop cultivation	(H. Li and Zhao, 2018)
Farmers shift to new water intensive crops or variety which demands more water	(Molle and Tanouti, 2017)
Rebound effect may occur by hydrological and economic mechanism	(Song et al., 2018)
Technology might increase the agricultural income through water intensive cropping and creating scarcity for others in a basin	(Pérez-Blanco et al., 2021)
Hydrological Mechanism (HM)	
HM is a combination of all the changes in the cropping system, crop change, increase in evapotranspiration because of technology	(Dagnino and Ward, 2012; Lecina et al., 2010; Song et al., 2018)
10–15 % higher evapotranspiration in almond crop cultivated by drip as compared to conventional methods	(Burt et al., 2001)
Drip irrigation increase yield which has positive correlation with evapotranspiration leading to HM	(Ward and Pulido-Velazquez, 2008)
Economic Mechanism (EM)	
The decision of a farmer is not motivated by water conservation, but by the economic gains from the MI adoption	(Brinegar and Ward, 2009)
EM includes shift to high water-consuming crops or crop variety, increased crop density, increase in area under cultivation	(Caswell et al., 1984; Song et al., 2018)
EM to the paradox can be extensive, i.e., more area	(Berbel and Mateos, 2014; Birkenholtz, 2017; Fishman et al., 2015; Molle and Tanouti, 2017; Pérez-Blanco et al., 2021)
EM to the paradox can be intensive, i.e., more dense cropping	(Contor and Taylor, 2013; Pfeiffer and Lin, 2014; Playán and Mateos, 2006; Ward and Pulido-Velazquez, 2008; Willardson et al., 1994)
Indian Studies	
No presence of paradox in hard rocks areas of Karnataka (India)	(Kabbur, 2017; Patil et al., 2015)
Absence of paradox based on no change in the proxy variables such as number of casing pipe and size of pump	(Bahinipati and Viswanathan, 2019)
A jump in the irrigated area of the farmers irrigated from 40 % to 67 %, indicating the paradox	(Birkenholtz, 2017)

(*continued on next page*)

3. Materials and methods

3.1. Sampling strategy

Micro-irrigation has been actively promoted among farmers by the Government of India through the *Pradhan Mantri Krishi Sinchayee Yojana* (Prime Minister's Agricultural Irrigation Scheme). Based on the information on micro-irrigation adoption available under this scheme, four major states of India — Uttar Pradesh (UP), Madhya Pradesh (MP), Maharashtra, and Telangana (see Fig. 1) — were selected for sampling. These states represent diversity in agro-climatic zones, cropping patterns, and the extent of micro-irrigation adoption. UP and MP exhibit a significant gap between potential and actual adoption of micro-irrigation (MI) technology. UP and Maharashtra have substantial sugarcane cultivation, which is highly water intensive (Garkar, 2017). Additionally, UP, MP, and Maharashtra are among the top three states in terms of the country's groundwater potential (CWC, 2019). Maharashtra and UP also experience among the highest annual rates of groundwater depletion (CWC, 2019). UP and MP showed lower MI adoption rates, while Maharashtra and Telangana exhibited higher adoption rates.

Fig. 2 outlines the purposive sampling strategy used to study micro-irrigation adoption and its impact. Within each state, two diverse districts were identified based on criteria such as micro-irrigation coverage, water availability, groundwater stress levels, and the presence of water-intensive crops like sugarcane, banana, soybean, and wheat. Similarly, two diverse blocks were selected within each district, followed by the identification of two diverse villages within each block. Subsequently, a random sampling strategy was employed in each village to select a sample that included both adopters and non-adopters of micro-irrigation across various land holdings. The paper utilised data from 404 adopter farmers (see Table 2).

3.2. Data

In agricultural operations, decisions and outcomes occur over extended periods, necessitating specific methods and instruments to capture information effectively. Panel data, involving data collection at different time points across seasons or years through multiple visits, is often considered a robust method to capture changes pre- and post-adoption of technologies such as micro-irrigation. However, it presents challenges such as the substantial time required to revisit farmers, high costs, logistical complexity, and sample attrition. Given the constraints of time and resources in our research project, we relied on retrospective recall data from farmers, especially regarding pre-adoption. Retrospective data, while convenient, is susceptible to

Table 1 (*continued*)

Key Highlights	Authors
In Andhra Pradesh (India), over three years, farmers switched to more profitable crops, sold saved water to neighbours, saw higher profits, but didn't notably reduce groundwater extraction	(Fishman et al., 2023)

Source: Authors' compilation

imperfect recall, making it relatively less reliable. Such data is subject to three types of errors: telescopic errors, where the timing of past events is misremembered; false negatives, where actual events or experiences are not reported; and false positives, where events or experiences that did not occur are reported (Coleman, 1983; Stewart et al., 1989; De Nicola and Giné, 2014). Despite these potential issues, the use of such data is frequent, including in fields such as game theory and decision science, where it helps understand changes and how past information/decisions relate to future decisions and outcomes (Piccione and Rubinstein, 1997; Battigalli, 1997).

Many studies in agriculture have utilized retrospective recall data (Berney and Blane, 1997; Waugh et al., 2009; Beegle et al., 2012; Rosinger, 2022; Adzawla et al., 2024). Beegle et al. (2012) demonstrated that recall data from farmers showed minimal significant bias, maintaining consistency whether collected immediately after harvest or eight months later. To mitigate the potential errors inherent in retrospective recall, strategies like triangulation, for example, using crop-cutting experiments to validate yield recall data (Adzawla et al., 2024), and specific interview techniques, can enhance accuracy (Berney and Blane, 1997). In our study, we applied triangulation by asking farmers about their increased area under micro-irrigation in two different ways: one inquiring about the overall cultivated area and another specific to crop area. This method helped cross-verify their responses and improve reliability. We also captured responses through detailed interviews with the farmers during the recall data collection.

Researchers developed a comprehensive survey tool, which was initially piloted in Gujarat and Maharashtra. After refining the survey schedule based on these pilot surveys, the finalized survey schedule was distributed to Agro-Economic Research Centers (AERC) for data collection. The survey schedule consisted of 13 sections covering, among other things, farmer and surveyor profiles, survey dates, location specifics, types of farmer respondents, demographic information, land use details, irrigation sources, cropping patterns, and pre- and post-adoption details and perception assessments, including satisfaction, willingness to expand technology adoption, challenges faced, and open-ended responses gauging their experiences. The data used in the paper covers two different cropping seasons—Kharif and Rabi—based on location and includes hours of irrigation as a proxy for water consumption. The data sample included 404 MI adopters across 86 villages, eight districts, and four states (see Table 2).

The survey was conducted with the support of professional staff from the four AERC of the Government of India, under the supervision of key

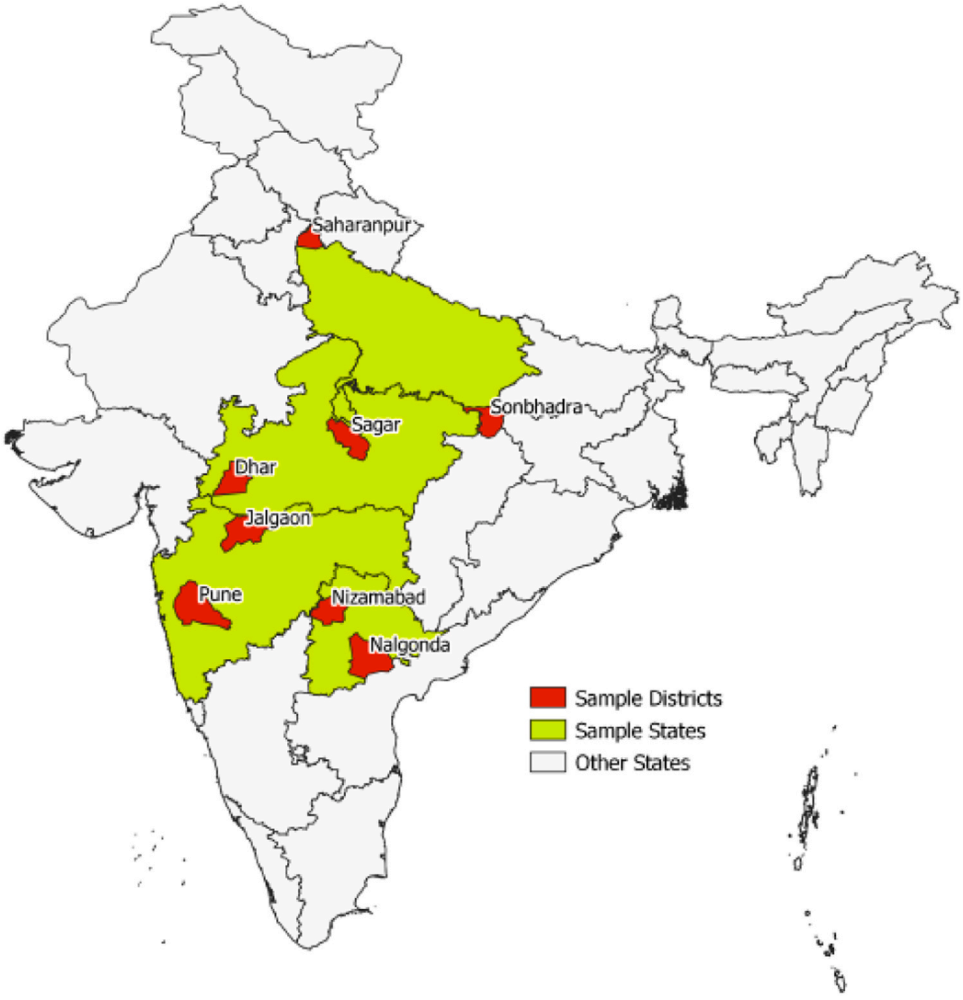


Fig. 1. Sampled States and Districts for the Study,.
Source: Authors' compilation

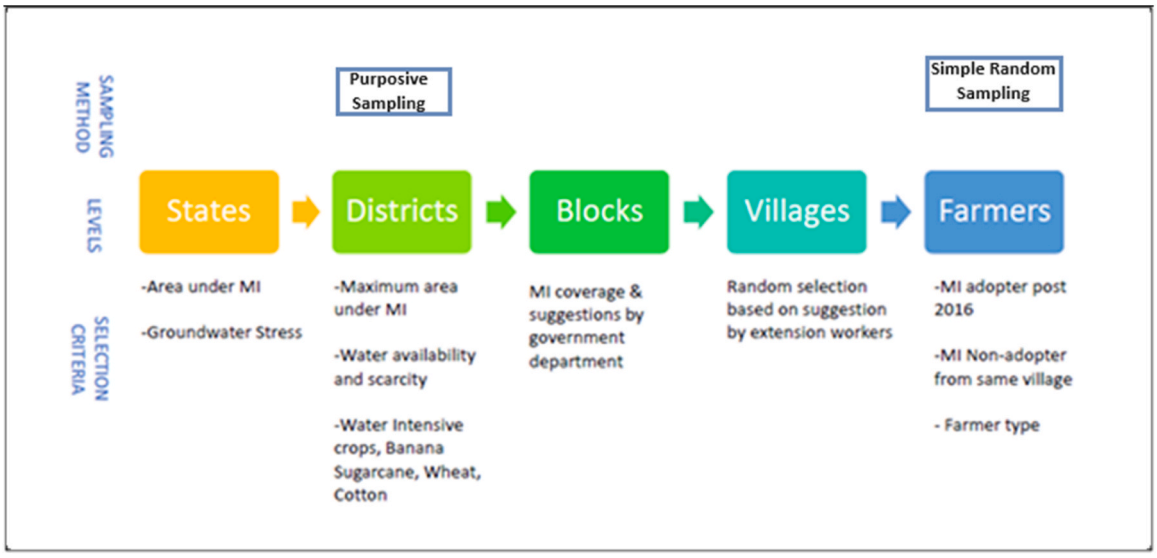


Fig. 2. Sampling Strategy.
Source: Authors' compilation

Table 2
Distribution of Sample Locations and Numbers.

Sr. No.	State	District	Villages	Samples
1	Uttar Pradesh	Sonbhadra	6	48
2	Uttar Pradesh	Saharanpur	7	48
3	Madhya Pradesh	Dhar	6	48
4	Madhya Pradesh	Sagar	17	48
5	Maharashtra	Pune	14	52
6	Maharashtra	Jalgaon	19	64
7	Telangana	Nizamabad	7	48
8	Telangana	Nalgonda	10	48
		Total	**86**	**404**

Source: Authors' compilation

researchers. It was administered from December 2019 to March 2020, encompassing cropping seasons from July 2018 to June 2019.

While acknowledging the limitations of the recall data, we utilized pre- and post-adoption information to evaluate changes in crop area, types cultivated, number of irrigations, and hours of pumping. The data included the area under various crops and the types of crops cultivated by farmers before and after the adoption of micro-irrigation. The analysis involved simple cross-tabulation to compare the status pre- and post-adoption of micro-irrigation. The paper also covers qualitative responses from farmers, many of which were captured on a scale from one to five. Enumerators explained the scale to respondents in their native languages.

Water used in irrigation can be measured in at least two ways: first, by using flow meters calibrated with the irrigation source (Goel, 2006); second, based on associated variables such as the horsepower of pump sets, water level in the well, efficiency of the pump, and duration of pumping (Rai and Mauria, 2006). The formula below can be used for calculating the quantity of water pumped or discharged during an irrigation:

$$V = \frac{P^{*}D}{9.8^{*}997^{*}H}{}^{*}1000 \quad (1)$$

V denote the volume of water discharged in liters, P represent the power of the irrigation pump in watts, D signify the duration of pumping for irrigation in seconds, and H denote the head length or water draft height from groundwater to the discharging pipe above the ground measured in meters. The values 9.8 and 997 correspond to the acceleration due to gravity and the conversion factor for mass to volume, respectively.

Due to limitations in time and resources, we were unable to conduct a thorough scientific calculation of the volumetric withdrawal of water, as it was impractical to install flow meters for the study's duration. However, previous studies (Narayanamoorthy, 2003; Shah et al., 2005) have utilized the pumping duration (D) of irrigation as a proxy for the extent of groundwater use in agriculture. Eq. 1 indicates that the duration of pump operation is directly proportional to the volume of water pumped. Therefore, we assume that greater hours of pumping indicate higher water usage.

We collected recall data from farmers regarding the hours of pumping for irrigation with micro-irrigation compared to traditional flood irrigation. Farmers were asked about the duration of irrigation for one crop cycle and the time required to irrigate their standard hectare of cropped land under both micro-irrigation and traditional methods. Multiplying the total number of irrigations by the hours taken per irrigation provided us with the total hours of irrigation in a crop cycle. This data was then used to estimate percentage changes in water use following the adoption of micro-irrigation.

3.3. Descriptive statistics

Table 3 presents key descriptive statistics from the data. On average, farmers operate 3.17 ha of land, with the average technology intensity—percentage area under the technology compared to total operated land—standing at 68 %. The cropping intensity averaged 1.41 before micro-irrigation and increased to 1.49 after adoption, indicating a rise in the gross area under crops. The Herfindahl index, a measure of crop concentration or diversification on the farm, averaged 0.581 before adoption and 0.559 post-adoption. A higher Herfindahl index signifies greater concentration (less diversification), suggesting an increase in crop diversification following MI adoption, as discussed in the analysis section. Other statistics include the average distance of 15 km to the nearest market from the farm, and an average household size of about six members. Farmers have, on average, about nine years of formal education, with an average age of 47 years.

4. Analysis and Results

This section presents the analysis and results. It begins by qualitatively and quantitatively examining the evidence of the extensive economic mechanism's impact following the adoption of micro-

Table 3
Summary of Descriptive Statistics.

Variable	Observations	Mean	Std. Dev	Min	Max
Technology Intensity (%)	404	68	29	4.80	100.00
Operated Area	404	3.165	3.086	0.41	26.32
Cropping intensity Before MI	404	1.406	0.488	0.33	4.00
Cropping intensity after MI	404	1.486	0.462	0.70	3.25
Herfindahl Index before MI	404	0.581	0.289	0.17	1.00
Herfindahl Index after MI	404	0.559	0.309	0.12	1.00
Market Distance	404	14.99	7.683	1.00	40.00
HH Size	404	5.75	2.515	1.00	19.00
Education Years	404	8.918	5.002	0.00	17.00
Age	404	47.12			

Source: Authors' compilation

irrigation—specifically, whether farmers increase their area under various crops. It then assesses the impact of the intensive economic mechanism, focusing on changes in crop types towards more water-intensive varieties alongside area increments due to micro-irrigation adoption. Subsequently, the section explores the evidence concerning changes in actual water use, measured in hours of pumping (a proxy), at the crop level to determine any positive or negative shifts in water consumption. Although a basin-level hydrological water budgeting approach would be optimal (Richter et al., 2017), this study focuses on farm and crop-level data analysis due to constraints in time and data availability. Lastly, the section discusses the implications of these findings for micro-irrigation policies to achieve the dual goals of water conservation and food security.

4.1. Expansion in cultivated area under micro irrigation

We investigated whether farmers increased their cultivated area following the adoption of micro-irrigation systems. Initially, we analysed farmer responses to two Likert scale questions. The first question assessed changes in overall cultivated area under micro-irrigation, with responses ranging from 1 ("Large Decrease") to 5 ("Large Increase"). The second question queried changes in area under specific crops post-adoption, also on a Likert scale from 1 ("Strongly Disagree") to 5 ("Strongly Agree"). Given multiple crops which a farmer cultivates, the n^c=770, where n^c is number of crops, and n^f=404 where n^f is number of adopter farmers sampled. We have presented odds ratio of the responses to the first question in the Table 4.

The results indicate that the odds ratio for "Large Decrease" is 0.004, suggesting a very low likelihood of farmers significantly reducing their crop area. An odds ratio of 0.036 for "Decrease" indicates a less frequent reduction compared to a large decrease but still uncommon. The "Neither" category has an odds ratio of 1.61, indicating a tendency towards neutral response or no change in area. The odds ratio of 0.36 for "Increase" indicates that farmers have a moderate likelihood of expanding their area under micro-irrigation. Lastly, "Large Increase" has an odds ratio of 0.086, indicating a lower probability of substantial expansion compared to modest increases. Overall, these results highlight a higher likelihood and frequency (61.7 %) of no change in cultivated area, with greater odds for an increase rather than a decrease.

Table 4
Frequency distribution of the perception responses and odds ratio on the question of change in area due to micro-irrigation adoption.

Area Change Perception	Freq.	Percent	Odds ratio
Large Decrease	3	0.4	0.004
Decrease	27	3.5	0.036
Neither	475	61.7	1.610
Increase	204	26.5	0.360
Large increase	61	7.9	0.086
Total	**770**	**100**	-

Source: Authors' compilation

In Fig. 3, we present the analysis of our second question, which investigates whether farmers increased the area under specific crops following the adoption of micro irrigation. The graph uses different colours to represent the frequencies of five categories of responses coded numerically from 1 to 5, capturing farmers' recollections ranging from Strongly Disagree to Strongly Agree. The two shades of green represent agreement, two shades of mustard indicate disagreement, and grey denotes a neutral response.

The graph indicates that among the sampled farmers (n^f = 404) cultivating specific crops (n^c =770), all farmers agreed that they increased the area under crops - strawberry, ridged gourd, papaya, *mosambi* (sweet lemon), cucumber, capsicum, and *ber* (jujube). A majority of farmers also responded in agreement for crops - beans (78 %), watermelon (70 %), maize (50 %), cowpea (50 %), bitter gourd (69 %), ginger (51 %), chickpea (50 %), tomato (43 %), chili (39 %), and banana (31 %). We observed that all these crops are high-value horticultural crops, with banana being particularly water-intensive. In our next analysis, we evaluated the changes in total and irrigated areas under all crops. Farmers whose total area increased with micro irrigation were coded as 1, while those who did not were coded as 0. We created a frequency distribution table for instances where the reported cultivated area had increased. Table 5 presents the state-wise and farmer category-wise frequency distribution of these farmers.

Farmer Category			
Marginal	68	75.0	75.0
Small	121	79.3	81.0
Medium	196	91.3	91.8
Large	19	94.7	100
Total	**404**	**85.1**	**86.1**

Source: Authors' compilation

The quantitative data clearly indicated to extensive mechanism of the rebound effect i.e., an increase in the overall cultivated area after micro-irrigation adoption. Quantitative data suggests that 85 per cent of the sampled farmers (n^c = 404) increased their total operated area under cultivation post adoption of MI. The highest percent increase in total operated area under cultivation was in Madhya Pradesh (97 %; n^f = 96), followed by Maharashtra (90 %; n^f = 116), Telangana (82 %; n^f = 96) and Uttar Pradesh (68 %; n^f = 96). Farmers in each category increased their area under irrigated agriculture. 94 % of the large farmers (n^f = 19) increased their total cultivated area followed by medium farmers (91 %; n^f = 196), and small (81 %; n^f = 121) and marginal (75 %; n^f = 68) farmers. In next sub-section we have presented detailed analysis on crop type and area changed under them.

4.2. Changes in crop choices under micro-irrigation

Many factors determine a farmer's choice of crop and cultivation method, including soil type and quality, water availability, market access and demand, government policies, labour availability, technological considerations, local context, and farmer experience (Amare et al., 2018; Birthal et al., 2020; Rahman, 2008). With the introduction of new irrigation technology, the question arises: what changes occur in the area under different crops and the crop mix? The previous section demonstrated an increase in operated area following the adoption of micro irrigation. In this section, we present data on farmers' crop types and their respective area changes across three categories: total operated area, irrigated area, and rainfed area. The sum of rainfed and irrigated areas constitutes the total area of a farmer's field. We begin by presenting the mean area change for some major crops post adoption of micro irrigation, followed by the absolute area change across these three

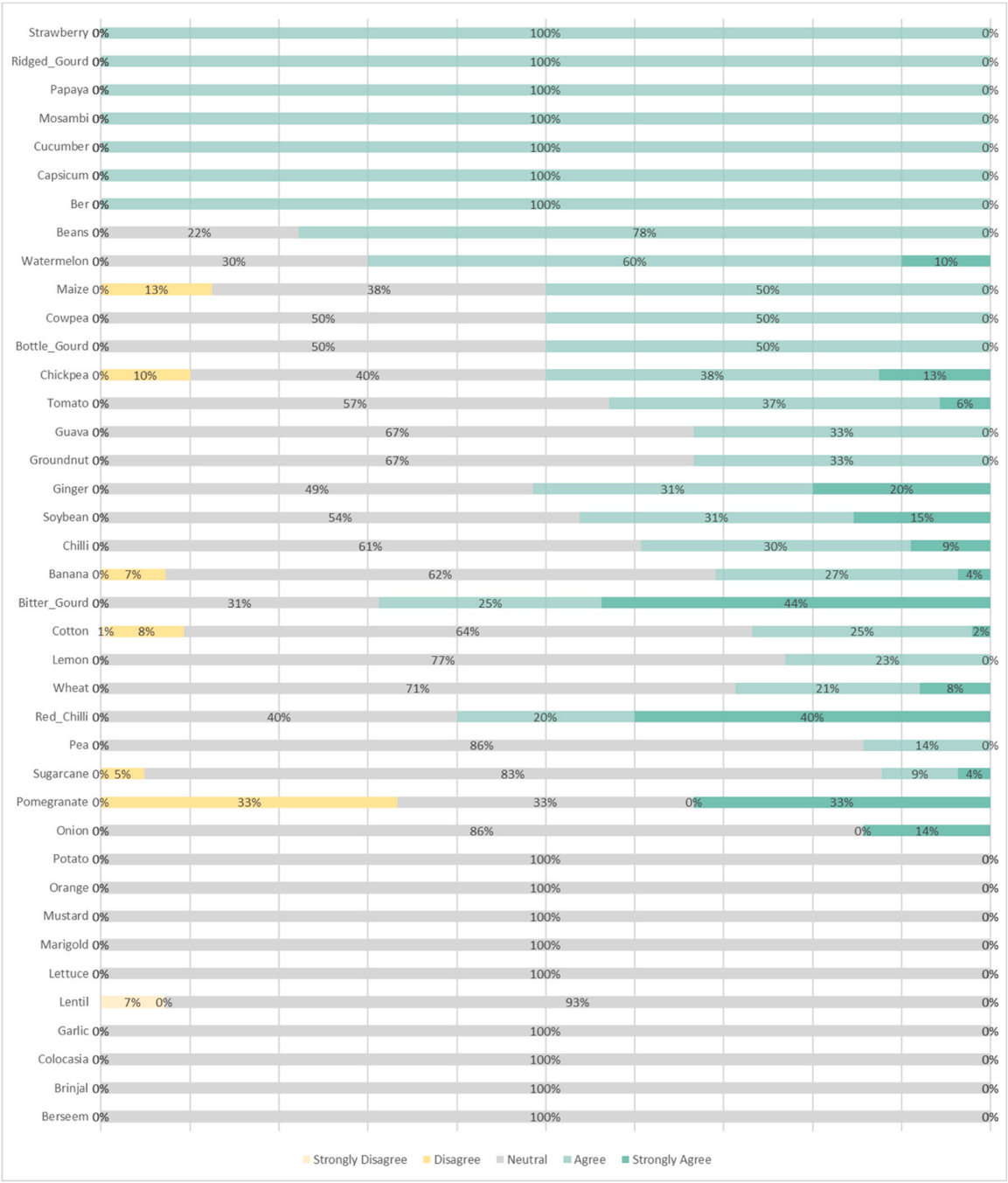


Fig. 3. Distribution of responses on change in area after adoption of micro-irrigation, sorted by crops.
Source: Authors' compilation

Table 5
Frequency Distribution of the farmers who increased area after Micro-Irrigation adoption.

Particulars		% Farmers	
	N	Total area	Irrigated area
	State		
Uttar Pradesh	96	68.8	68.8
Madhya Pradesh	96	97.9	97.9
Maharashtra	116	90.5	90.5
Telangana	96	82.3	86.5

categories. Table 6 presents the mean area and percentage change in mean area under major crops before and after the adoption of micro irrigation.

The mean cultivated areas under sugarcane and wheat increased by approximately six percent after adopting micro-irrigation. Banana, a water-intensive crop, experienced an 86 % increase in mean cultivated area post-adoption of micro-irrigation. The second-highest mean operated area increase was observed among ginger farmers (81 %), followed by soybean (71 %) and chili (60 %). The only crop that saw a reduction in cultivated area post-adoption was cotton, with a mean operated area decrease of 25 %. During farmer group discussions in the survey, banana farmers reported that they increased their cultivated area substantially following micro-irrigation adoption. For example, in an interview, a

Table 6
Change in area after adoption of Micro Irrigation (MI) in major crops.

Crop	Observations	Mean Area in hectare		%Change
		Before MI	After MI	
Wheat	99	1.438	1.529	6.3
Sugarcane	82	1.428	1.514	6.0
Chickpea	71	1.966	2.381	21.1
Banana	50	1.572	2.931	86.5
Cotton	68	2.494	1.872	−24.9
Soybean	38	2.049	3.499	70.7
Chili	42	0.415	0.664	59.9
Ginger	31	0.286	0.518	81.1

Source: Authors' compilation

large farmer, owning 80 ha land in Jalgaon district of Maharashtra mentioned digging eight more bore-wells on his farm. He expanded the cultivated area under banana after realizing increased yields and income from micro-irrigation. Table 7 presents the absolute values of the changes in crop area after adopting micro-irrigation.

The table indicates that farmers reduced their area under crops such as paddy, wheat, maize, and jowar. Additionally, the cultivated area under rainfed conditions has decreased, with a significant portion of this reduction specifically involving pulses such as soybean, urd, and mung bean. These crops have shown an increase in their irrigated area. As the irrigated area increased, farmers reduced the total operated area under cereals and the rainfed area under pulses. We have categorised the crop data from Table 7 into commercial crops, cereals, pulses, vegetables, oilseeds, and fruit crops, and the results are presented in Table 8.

Table 7
Crop Profile Change after the adoption of Micro-irrigation (MI).

Crop Type	Total Area (in Ha)			Irrigated Area (in Ha)			Rainfed Area (in Ha)		
	Before MI	After MI	Change	Before MI	After MI	Change	Before MI	After MI	Change
Soybean	233.6	253.9	20.2	218.9	253.4	34.5	14.7	0.5	−14.2
Chickpea	171.9	222.8	51.0	152.2	221.0	68.8	19.7	1.8	−17.9
Wheat	259.8	219.9	−39.9	259.8	219.9	−39.9	-	-	-
Paddy	224.8	209.0	−15.8	224.8	207.8	−17.0	-	1.2	1.2
Banana	116.0	162.8	46.7	116.0	162.8	46.7	-	-	-
Cotton	336.1	300.5	−35.7	275.1	257.4	−17.8	61.0	43.1	−17.9
Sugarcane	123.0	133.7	10.7	123.0	131.1	8.2	-	2.6	2.6
Maize	62.5	60.3	−2.2	52.9	49.9	−3.0	9.6	10.4	0.8
Urad	39.4	44.3	4.9	-	43.5	43.5	39.4	0.8	−38.6
Chili	27.1	36.9	9.9	26.9	36.9	10.1	0.2	-	−0.2
Tomato	25.0	35.1	10.1	25.0	35.9	10.8	-	−0.8	−0.8
Ginger	11.3	18.7	7.4	11.3	18.7	7.4	-	-	-
Lemon	-	16.3	16.3	-	16.3	16.3	-	-	-
Groundnut	0.8	15.8	15.0	0.8	17.8	17.0	-	−2.0	−2.0
Watermelon	1.2	13.5	12.3	1.2	13.5	12.3	-	-	-
Bitter Gourd	5.7	11.1	5.4	5.7	11.1	5.4	-	-	-
Onion	11.7	11.1	−0.6	11.7	11.1	−0.6	-	-	-
Lentil	-	9.7	9.7	-	8.8	8.8	-	0.8	0.8
Maize intercrop	9.1	9.1	-	9.1	9.1	-	-	-	-
Red Chili	-	7.7	7.7	-	7.7	7.7	-	-	-
Orange	7.5	7.5	-	7.5	7.5	-	-	-	-
Bottle Gourd	1.2	6.4	5.2	1.2	6.4	5.2	-	-	-
Mustard	3.4	6.0	2.6	3.4	6.0	2.6	-	-	-
Vegetable	2.8	5.9	3.0	2.8	5.9	3.0	-	-	-
Guava	-	5.3	5.3	-	5.3	5.3	-	-	-
Pea	3.5	5.0	1.5	3.5	5.0	1.5	-	-	-
Fodder	3.2	4.2	1.0	2.4	3.4	1.0	0.8	0.8	-
Berseem	2.1	4.2	2.1	2.1	4.2	2.1	-	-	-
Horse Gram	4.9	4.1	−0.8	4.9	4.1	−0.8	-	-	-
Garlic	-	4.0	4.0	-	4.0	4.0	-	-	-
Beans	0.2	3.8	3.6	0.2	3.8	3.6	-	-	-
Arhar	1.5	3.6	2.1	-	0.8	0.8	1.5	2.8	1.3
Red gram	0.8	3.2	2.4	-	2.4	2.4	0.8	0.8	-
Moong	-	3.0	3.0	-	3.0	3.0	-	-	-
Pomegranate	0.6	3.0	2.4	0.6	3.0	2.4	-	-	-
Jowar	13.4	3.0	−10.4	8.6	0.6	−8.0	4.7	2.4	−2.3
Ber	-	2.9	2.9	-	2.9	2.9	-	-	-
Papaya	-	2.4	2.4	-	2.4	2.4	-	-	-
Marigold	-	2.3	2.3	-	2.3	2.3	-	-	-
Lettuce	-	2.0	2.0	-	2.0	2.0	-	-	-
Potato	0.8	2.0	1.3	0.8	2.0	1.3	-	-	-
Ridged Gourd	-	1.8	1.8	-	1.8	1.8	-	-	-
Sunflower	-	1.6	1.6	-	1.6	1.6	-	-	-
Moambe	0.8	1.6	0.8	0.8	1.6	0.8	-	-	-
Sesamum	0.3	1.5	1.2	-	1.2	1.2	0.3	0.3	-
Cowpea	0.6	1.4	0.8	0.6	1.4	0.8	-	-	-
Colocasia	1.6	1.2	−0.4	1.6	1.2	−0.4	-	-	-
Capsicum	1.1	1.1	0.0	1.1	1.1	0.0	-	-	-
Brinjal	0.8	1.0	0.2	0.8	1.0	0.2	-	-	-
Cucumber	-	1.0	1.0	-	1.0	1.0	-	-	-
Strawberry	-	0.8	0.8	-	0.8	0.8	-	-	-
Bajra	0.5	0.6	0.1	-	0.6	0.6	0.5	-	−0.5
Rose	-	0.3	0.3	-	0.3	0.3	-	-	-
Total	**1710.6**	**1889.9**	**179.3**	**1557.4**	**1824.4**	**267.0**	**153.2**	**65.5**	**−87.7**

Source: Authors' compilation

There has been an increase in the total operated area of sampled farmers by 179 ha (10 %) after the adoption of micro-irrigation. Farmers brought an additional 267 ha (17 %) of land under irrigated agriculture. There has been a decline in rainfed area by 87 ha, a 57 % reduction. The major increase in crop area from the baseline is in oilseeds (451 %), horticulture (73 %), vegetables (65 %), and pulses (22 %). It is notable that the area under cereal crops declined by 68 ha (12 %). In irrigated area, an additional 167 ha came under pulses, 92 ha under horticulture, 58 ha under vegetables, and 22 ha under oilseeds. Rainfed area under pulses declined by 68 ha, which has been converted into irrigated area. Fig. 4 presents the changes in the area of crop categories in percentage.

In Fig. 4, we observe that the percentage area under cereals and commercial crops such as sugarcane and cotton decreased. Conversely, there was an increase in the area under pulses, vegetables, fodder, and horticultural crops following the adoption of micro-irrigation. There was also a decline in the rainfed area under pulses. The results from our analysis confirm that farmers shifted their crop choices from cereals to more lucrative horticulture crops. We further examine how the area has changed for selected water-intensive crops—sugarcane, banana, chili, tomato, soybean, and groundnut, as identified by Rai and Mauria (2006).

The total operated area under all water-intensive crops increased by 76.9 ha (9 %), while the irrigated area increased by 109 ha (14 %) and rainfed area declined by 32 ha (43 %). All crops, except cotton, saw an increase in both total and irrigated area. The highest increase in total area was observed in banana (47 ha), followed by soybean (20 ha) and groundnut (14 ha). In percentage terms, these increases were 41 %, 9 %, and 1863 %, respectively. Soybean farmers brought a substantial additional area under irrigation (16 %) following the adoption of micro-irrigation, which was previously rainfed. Farmers who reduced or stopped cotton cultivation mostly hailed from Madhya Pradesh and Telangana, shifting to crops like soybean, ridged gourd, cowpea, and bottle gourd. These farmers used to cultivate cotton using flood irrigation and some traditional varieties under rainfed conditions. It is noteworthy that cotton is a longer-duration crop and poses higher risks. Investments in micro-irrigation prompted farmers to opt for more lucrative and shorter-duration crops such as soybean and vegetables. Micro-irrigation adoption also had a positive change on crop diversification, as measured by the Herfindahl index presented earlier in descriptive statistics, though we are not indicating the attribution solely to micro irrigation. In addition to the crops analysed here, we present the list of all new crops farmers added post adoption of micro-irrigation in Table 10.

The 15 new crops mentioned above were added by farmers on 61 ha of land after investing in micro-irrigation. The largest area was dedicated to lemon, followed by lentil and red chili. Farmers predominantly adopted new crops in the horticultural category, including vegetables and pulses. We now shift our focus to the changes in irrigated area by crop with micro-irrigation. The results are presented in Table 11.

The highest increase in irrigated area is seen in chickpea (68.8 ha), followed by banana (46.7 ha), groundnut (17 ha), watermelon (12.3 ha), tomato (10.8 ha), chili (10 ha), and sugarcane (8 ha). This is of concern since most of the crops are water-intensive. The highest percentage increase was observed in groundnut, watermelon, beans, and pomegranate—all high-value crops. While we acknowledge that the selection of crops following the adoption of micro-irrigation cannot be exclusively attributed to the technology—considering other influencing factors such as rainfall, market access, and government policy—we posit a reasonable assumption that farmers tend to choose more lucrative crops under micro-irrigation systems. In the next section, we present results on changes in water use by different crops using the proxy of hours of pumping for irrigation.

4.3. Assessment of change in use of water

Assessing the change in water use with the adoption of micro-irrigation technology is crucial to this study. We used hours of groundwater pumping as a proxy variable for the quantity of water extracted and used by farmers. We conducted the analysis using simple linear regression, with the total hours of pumping per hectare as the dependent variable and adoption of micro-irrigation as the independent variable, to assess the change and its statistical significance. The results are presented in Table 12 below and visually depicted in Fig. 5. The findings indicate that for all crops, there is a significant decrease in the total hours of water pumping per hectare after the adoption of micro-irrigation. The highest decrease in hours of irrigation is observed in cotton (52 %), followed by wheat (41 %), and banana and sugarcane (39 %). Overall, across all crops, there is a significant reduction of 32 % per hectare.

The evidence suggests that for all crops, micro-irrigation significantly reduces the hours of pumping compared to conventional irrigation. While this reduction may not be entirely attributable to the adoption of micro-irrigation, it is certainly a major contributing factor. The water use efficiency of micro-irrigation is well established in the literature. However, other factors such as reduced weed population, rainfall before and after adoption, and lower evaporation losses during irrigation also play a role in this phenomenon.

We further examined the change in absolute hours of pumping for all cropped areas (including any changes in area—not per hectare) with the adoption of micro-irrigation. We present results only for those crops which show a positive change (increase) in absolute hours of pumping for irrigation. This calculation includes hours of pumping before and after adoption of micro-irrigation for a specific crop, considering changes in the area under the crop. Table 13 presents the results for crops which show an increase in the hours of pumping.

Results show that in chickpea, ginger, soybean, and beans, there is an increase in absolute hours of pumping after the adoption of micro-irrigation. This increase is highest in chickpea at 55 %, followed by bottle gourd (27 %) and ginger (20.9 %). For these crops, we observe an absolute increase in water use by the farmers. The results indicate that farmers saved water per unit area across all crops, but these specific crops showed an increase in volumetric water use after MI adoption.

Table 8
Crop category wise area change post adoption of Micro-Irrigation (MI).

Crop Type	Total Area (Ha)			Irrigated Area (Ha)			Rainfed Area (Ha)		
	Before MI	After MI	Change	Before MI	After MI	Change	Before MI	After MI	Change
Cereals	570.1	501.9	−68.2	555.2	487.9	−67.3	14.8	14.0	−0.8
Commercial	459.1	434.2	−24.9	398.1	388.5	−9.6	61.0	45.7	−15.3
Fodder	5.3	8.4	3.1	4.5	7.5	3.1	0.8	0.8	0.0
Horticulture	126.2	218.7	92.5	126.2	218.7	92.5	0.0	0.0	0.0
Oilseeds	4.5	24.8	20.3	4.2	26.6	22.4	0.3	−1.8	−2.0
Pulses	456.4	554.8	98.4	380.3	547.3	167.0	76.1	7.5	−68.6
Vegetables	89.2	147.2	58.0	88.9	147.9	58.9	0.2	−0.8	−1.0
Total	**1710.6**	**1889.9**	**179.3**	**1557.4**	**1824.4**	**267.1**	**153.2**	**65.5**	**−87.7**

Source: Authors' compilation

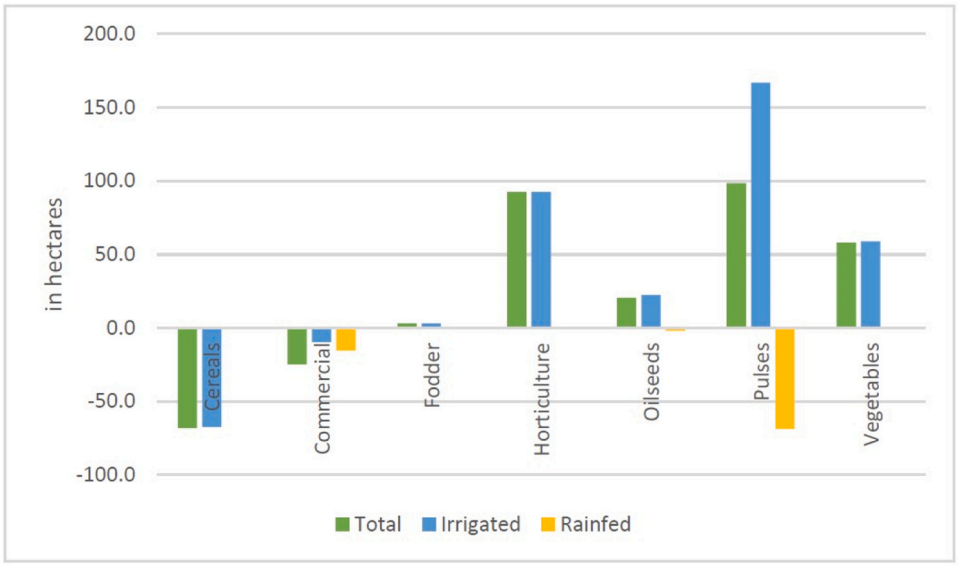


Fig. 4. Change in the crop area post adoption of micro-irrigation in hectares.
Source: Author's compilation

Table 9
Change in area in water intensive crops.

Crop Type	Total Area			Irrigated Area			Rainfed Area		
	Before MI	After MI	Change	Before MI	After MI	Change	Before MI	After MI	Change
Cotton	336.1	300.47	−35.7	275.1	257.4	−17.8	60.9	43.1	−17.9
Sugarcane	122.9	133.7	10.7	122.9	131.1	8.2	-	2.6	2.7
Banana	116.0	162.8	46.7	116.0	162.8	46.7	-	-	-
Groundnut	0.8	15.8	14.9	0.8	17.8	16.9	-	−2.0	−2.0
Soybean	233.6	253.9	20.2	218.9	253.4	34.5	14.7	0.5	−14.2
Chili	27.1	36.9	9.8	26.9	36.9	10.1	0.2	-	−0.2
Tomato	25.0	35.1	10.1	25.0	35.9	10.8	-	−0.8	−0.8
Total	**861.7**	**938.6**	**76.9**	**785.7**	**895.2**	**109.4**	**75.9**	**43.4**	**−32.6**

Source: Author's compilation

Table 10
Area under new crops post adoption of Micro Irrigation.

New Crops	Area (ha)
Lemon	16.31
Lentil	9.65
Red Chili	7.69
Guava	5.26
Garlic	3.97
Moong	3.04
Ber	2.94
Papaya	2.43
Marigold	2.25
Lettuce	2.02
Ridged Gourd	1.83
Sunflower	1.62
Cucumber	1.01
Strawberry	0.79
Rose	0.26
Total	**61.07**

Source: Author's compilation

5. Discussion

5.1. Presence of the economic mechanism of the paradox

Literature indicates that the water saved by micro-irrigation may be used to extend the area under irrigation (Song et al., 2018), a concept termed as extensive economic mechanism. How this saved water will be used depends on the demand elasticity of water (Caswell et al., 1984; Dumont et al., 2013). The decision to expand the farm area under micro-irrigation depends on factors such as the amount of unused land a farmer has, the additional income farmers can generate with the new technology, and the availability and stress on labour resources (Lecina et al., 2010; Pfeiffer and Lin, 2014). Our results indicate that farmers increased their cultivated area after adopting micro-irrigation, with over 65 % of farmers across all states expanding their operated area post MI adoption. The highest increases were seen in Madhya Pradesh (98 %), followed by Maharashtra (90 %), Telangana (82 %), and Uttar Pradesh (68 %). Although a limited number of farmers initially reported an increase in area to our first question, a detailed analysis of crop-wise and

Table 11
Crops Experiencing Increase in the Irrigated Area Post Adoption of Micro-Irrigation.

Irrigated Area under Crops				
Crop	**Before MI**	**After MI**	**Change**	**%Change**
Groundnut	0.81	17.79	16.98	2098
Beans	0.21	3.82	3.61	1719
Watermelon	1.21	13.54	12.33	1016
Bottle Gourd	1.21	6.43	5.22	431
Pomegranate	0.61	3.04	2.43	400
Potato	0.75	2.02	1.27	169
Fodder	4.45	7.55	3.1	144
Cowpea	0.6	1.39	0.79	132
Other Vegetables	2.84	5.87	3.03	107
Mosambi	0.81	1.62	0.81	100
Bitter Gourd	5.7	11.12	5.42	95
Mustard	3.4	5.95	2.55	75
Ginger	11.34	18.73	7.39	65
Chickpea	152.19	221.01	68.82	45
Pea	3.5	5.02	1.52	43
Tomato	25.02	35.85	10.83	43
Banana	116.02	162.76	46.74	40
Chili	26.87	36.92	10.05	37
Brinjal	0.81	1.02	0.21	26
Sugarcane	122.97	131.12	8.15	7
Capsicum	1.09	1.13	0.04	3
Total	**482.41**	**693.69**	**211.28**	**44**

Source: Author's compilation Red marked are water intensive crops

Table 12
Regression estimates of total hours of pumping in irrigation and percentage change across key crops.

Crops		**Coefficients**		**% change**
		Constant	**MI Adoption**	
Wheat (N=198)	Coefficient	218.3	−89.68	−41.08
	t-Stat	14.09	−3.87	
	Significant	***	***	
Sugarcane (N=164)	Coefficient	218.3	−84.86	−38.87
	t-Stat	14.09	−3.87	
	Significant	***	***	
Cotton (N=121)	Coefficient	28.6395	−15.15	−52.91
	t-Stat	19.47	−7.72	
	Significant	***	***	
Banana(N=100)	Coefficient	392.12	−153.75	−39.21
	t-Stat	10.87	−3.01	
	Significant	***	***	
All Crops (N = 1051)	Coefficient	116.73	−37.45	−32.08
	t-Stat	22.9	−5.31	
	Significant	***	***	

* p<0.1
** p<0.05,
*** p<0.01,
Source: Author's compilation

area data confirmed the mechanism.

Farmers in India are increasingly facing two major economic challenges in their agricultural decisions. Firstly, declining returns in traditional crops such as rice, wheat, and cotton due to poor real prices and rising input costs of irrigation, fertilisers, and labour (Rajkhowa and Kubik, 2021). Secondly, there is a scarcity of available labour. Given these challenges, crop diversification and increasing crop diversity are crucial strategies to improve returns and mitigate risks. A portfolio of crops with low correlated returns helps distribute risks among different crops (Birthal et al., 2019). In our study, farmers shifted towards diverse and lucrative horticultural and vegetable crops, increased the area under irrigated pulses, and reduced acreage under cereals and cotton, thereby improving the Herfindahl index. These results are consistent with findings by Fishman et al. (2023) in Andhra Pradesh, where farmers turned towards horticulture and vegetables for higher returns, and surplus saved water was sold to neighbouring farmers. The sampled farmers added approximately 15 new horticultural crops to their portfolios and converted 87 ha of rainfed land to irrigated agriculture. More than 75 % of small and marginal farmers were able to irrigate more crop areas than before adopting micro-irrigation.

5.2. Water use per crop

To better examine the extent of water saved, we used a proxy of water volume use, i.e., hours of pumping. We attempted to observe the per-hectare saving of water post-adoption of MI through the hours of pumping. Results indicated that farmers reduced pumping hours by 41 % in wheat, 38 % in sugarcane, 52 % in cotton, 39 % in banana, and 32 % overall across crops. The analysis suggests that the technology reduces the hours of pumping for irrigation and thus reduces per-unit area water consumption. We found, however, that with an increase in area, there was an absolute increase in pumping hours by farmers in some crops chickpea, ginger, soybean, watermelon, beans, and bottle gourd. The empirical evidence substantiates the assertion that micro-irrigation enhances water use efficiency across a broad spectrum of crops, a finding also corroborated by existing scholarly work (Bhamoriya and Mathew, 2014; Kumar, 2016; J. Li et al., 2018; Palanisami et al., 2011). Farmers' ex-ante adoption behaviour regarding the transference of per-crop water savings becomes pertinent to understand. Our study indicates farmers exhibited an increasein groundwater extraction (measured by irrigation hours) in a few select crops through increased area, including in some water-intensive crops and changing their cropping pattern. Consequently, the ultimate impact of micro-irrigation adoption on water use and conservation hinges on the allocation of the saved water by farmers—whether it is conserved or directed towards expanding cultivated areas, transitioning to more water-intensive crops, selling the conserved water, or additional non-farm activities such as animal husbandry. The incidence of the Jevons paradox would depend on these factors.

5.3. Policy implications

An increase in irrigated area and enhanced water use efficiency is an encouraging outcome for the mandate *'Har Khet Ko Pani'* (water for every farmland) under the *Pradhan Mantri Krishi Sinchayee Yojana* policy of the Government of India (GoI, 2020). However, this comes at the cost of sustainable food security, a concern highlighted in research studies, see Fishman et al., (2023); Pérez-Blanco et al., (2021), which emphasize groundwater irrigation limitations. Relying solely on micro-irrigation (MI) technology to address water conservation and sustainable production in agriculture is insufficient. Effective water conservation requires regulatory interventions that prioritize all - farmers' quality of life, rural development, and sustainable groundwater use. This objective can be achieved through public policy interventions that also consider farmers' behavioural responses to subsidies aimed at improving irrigation efficiency. While increasing irrigation efficiency is crucial, it must be complemented by robust water accounting at the basin level, limiting extractions, assessing uncertainties, valuing trade-offs, and understanding irrigators' incentives. Additional policy suggestions include considering crop-based subsidies and support prices based on agro-ecological stress. For instance, in Punjab and Haryana states of India, currently facing extreme water stress (Singh et al., 2021) crops such as rice should be phased out from the region, as micro-irrigation may not sufficiently address the hydrological challenges. In line with classical economics, farmers aim to maximize benefits by using saved water from micro-irrigation to enhance returns from the operated land. Therefore, it is crucial to develop market strategies that encourage the cultivation of low water-intensive, high-return crops like fruits and vegetables to prevent undesirable consequences without jeopardising farmer households' welfare. The research emphasizes that the success of MI adoption in addressing water challenges in agriculture requires a

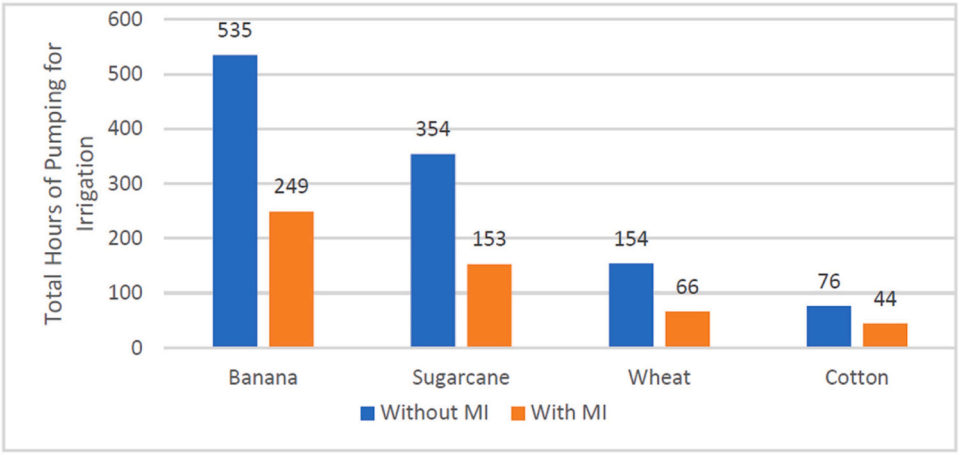


Fig. 5. Difference in mean total hours of pumping in three water intensive crops.
Source: Author's compilation

Table 13
Increase in Absolute Hours of Pumping after adoption of Micro-Irrigation.

Crop	Before	After	Change	%
Chickpea	2866	4458	1592	55.5
Ginger	6475	7830	1355	20.9
Soybean	1600	1862.5	262.5	16.4
Beans	562	650	88	15.7
Bottle Gourd	165	210	45	27.3

Source: Authors' compilation

holistic approach—robust institutional measures, a policy paradigm shift, and a deeper understanding of farmers' behaviours.

5.4. Limitations and extension of study

The study acknowledges its reliance on recall data, recognising its inherent limitations. While the optimal methods for measuring impact involve either a difference-in-differences approach or a randomised control trial, the constraints of time and resources necessitated the utilisation of recall data for a pre-post scenario analysis of farmers. Despite these limitations, the findings serve as an impetus for further research, particularly within a more robust policy framework. Given the data limitations, it would be imprudent to attribute the increase in agricultural area and changes in crop cultivation solely to micro-irrigation. Crop choices depend on numerous other factors, including enhanced institutional access to markets, credit, subsidies, influence from neighbouring farmers, and participation in collectives (Amare et al., 2018; Birthal et al., 2020; Rahman, 2008). We acknowledge the crop changes post the adoption of micro-irrigation and suggest exploring the research in a more robust manner. Future studies might consider hydrogeological approaches, assessing the treatment effect of introducing micro-irrigation in a basin/watershed and the behavioural responses of farmers to the investment in micro-irrigation technology.

6. Conclusion

The study indicates the rebound effect of micro-irrigation adoption. MI technology contributes significantly to water use efficiency in crops; the research suggests that relying solely on the technology may not be sufficient for sustainable agriculture. Farmers increased their crop area under micro-irrigation and adopted more diverse, remunerative and water intensive horticultural crops. The analysis of water use efficiency revealed substantial savings in hours of pumping, indicating a positive impact on water consumption per unit area. However, some crops, including chickpea, ginger, soybean, watermelon, beans, and bottle gourd, experienced an absolute increase in water pumping hours, indicating a rebound effect. We encourage further research on actual water use by adopting hydrogeological approaches, basin-scale water accounting, installation of flow meters to calculate actual consumptive use, and analysis of farmers' behavioural responses pre-adoption.

Author contribution

Gurpreet Singh and Vasant P. Gandhi contributed to the study's conception and design. Gurpreet Singh was responsible for material preparation, data collection, analysis, and writing the first draft. Vasant P. Gandhi provided mentorship and guidance in refining the drafts, while Dinesh Jain contributed to final editing and feedback. All authors have read and approved the final manuscript.

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CRediT authorship contribution statement

Gurpreet Singh: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vasant Gandhi:** Writing – review & editing, Supervision, Funding acquisition. **Dinesh Jain:** Writing – review & editing.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Language Tool an AI based online grammar, style & spell checker tool to improve the readability and ensure coherence. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of Competing Interest

The authors declare that we are the sole author of this manuscript entitled "**Micro-irrigation adoption and indication to Jevons' Paradox: A case from four states of India**" and that we have no conflicts of interest regarding the research, authorship, and/or publication

of this article. To the best of our knowledge, the research reported in this manuscript has been conducted in accordance with the guidelines of the institution(s) where the research was conducted. Furthermore, the information and data in this manuscript is accurate and has not been submitted for publication elsewhere. The research was supported by government funding, given to an independent institution Centre for Management in Agriculture at the Indian Institute of Management Ahmedabad, to conduct the research. The findings are not at all influenced by the government agency or stakeholders.

Data Availability

Data will be made available on request.

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